Adaptive Trial Design for the RAS-Inhibitor Era

How can we evaluate hundreds of RAS-directed trial designs before patients are enrolled?

Naim Rashid, PhD

Today's argument

> **New RAS therapies need new kinds of trials.**

Methods to optimize combination doses are established. Designs that track resistance and adapt the trial in response are enrolling.

What's missing is making those designs practical to calibrate and run — **the gap BATON is built to close.**

The argument of the talk. The science of RAS-era combinations puts real demands on trial design — combination dose-finding under non-monotonic dose-response, and resistance tracking with in-trial adaptation. Modern Bayesian adaptive designs answer both, at different maturity levels: dose-optimization methods are established, resistance-tracking designs are enrolling. What's been missing is the infrastructure to make them practical, and that's where BATON closes the gap. BATON is the closing tool, not the headline. Note the deliberate asymmetry in the fragment — "established" vs "enrolling" — this prevents the resistance-side overclaim that would land badly with a SPORE audience who knows EVOLVE hasn't yet reported.

The clinical case

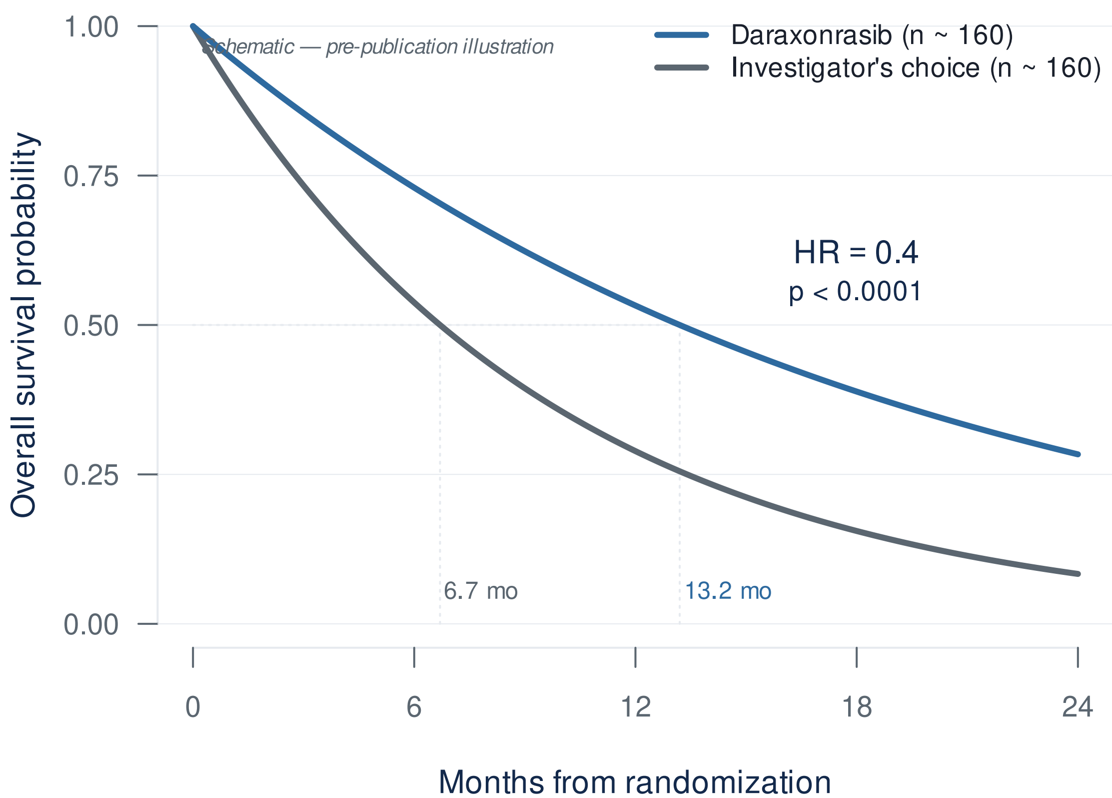
Why combinations, and why now?

Three slides on the clinical pressure before we get to design problems.

The RAS revolution has arrived for PDAC

- Daraxonrasib (RMC-6236): oral, multi-selective RAS(ON) inhibitor
- **RASolute 302** (2L mPDAC)[1]:
  - **Median OS: 13.2 vs 6.7 months**
  - **HR 0.40 — 60% reduction in risk of death** ($p < 0.0001$)
- ASCO 2026 Plenary, May 31 (Wolpin); *NEJM* same day

O'Reilly et al., *NEJM* 2026 (RASolute 302) · ASCO Plenary Abstr LBA5



For pancreatic cancer this is the largest survival signal in a generation. Final analysis was presented at the ASCO Plenary on May 31 and published in *NEJM* the same day. Cite the NEJM paper, not a press release — this room will care about that. Two numbers to have in your pocket for Q&A. Median follow-up was 8.5 months at the February 10 data cutoff, so if anyone presses on maturity that's your number. And the 13.2 vs 6.7 figure is overall ITT; the dual primary endpoint was OS and PFS in the RAS G12-mutant subgroup (228 vs 231 patients), where HR was also 0.40 (P = 5.9e-10). If a discussant draws the ITT-vs-primary distinction, the answer is that the 0.40 benefit held in both — robustness across RAS mutation status is the whole point of a multi-selective RAS(ON) inhibitor. That robustness is a strength to lean on, not a vulnerability. Whatever else you take from this talk, this result will reshape the GI oncology trial landscape over the next two to three years — every program in this room will soon be deciding what to combine with this drug, in which biomarker-defined patients.

…but monotherapy has visible ceilings

RMC-6236-001 (NEJM 2026)[2]

Full cohort (n=168 PDAC, doses 300 mg):

Metric	Value
Grade 3 TRAEs	**30%**

At 300 mg (the Phase 3 monotherapy dose):

Metric	Value
Grade 3 TRAEs	~**34%**
Required dose modification	**48%**
TRAE-related discontinuation	**0%**

48% is the headroom number. Every combination partner you add is dose-modification room you no longer have.

Wolpin et al., *NEJM* 2026 (RMC-6236-001 PDAC cohort)

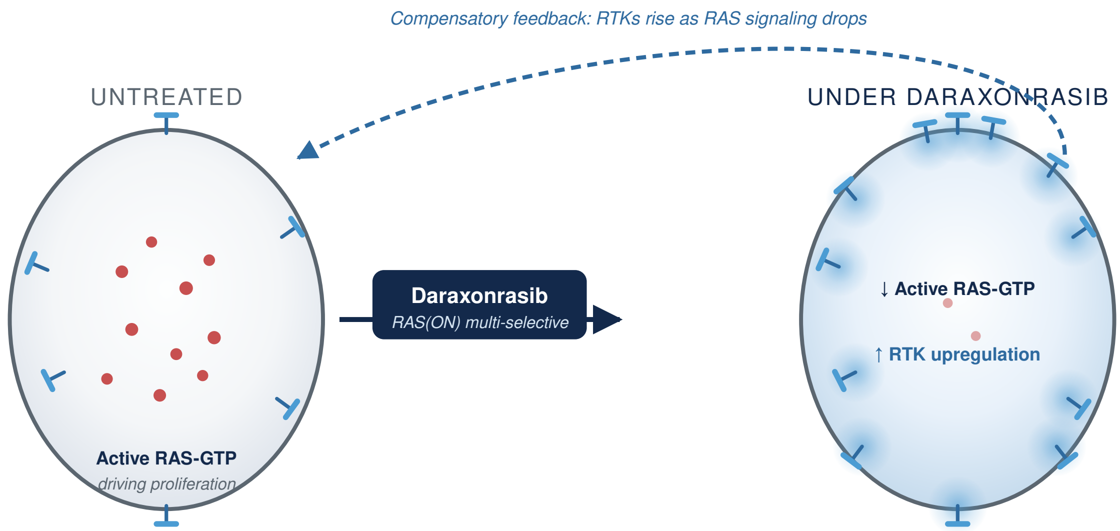
Three numbers worth keeping in mind. Across the full cohort of 168 patients on doses up to 300 mg, about a third have Grade 3 or worse toxicity. No patient discontinued because of drug toxicity — that's genuinely remarkable for an active drug in this disease. The qualifier matters; everyone with metastatic PDAC eventually comes off for progression, but nobody had to come off this drug for tolerability. Now narrow to the 300 mg dose specifically, which is the Phase 3 monotherapy dose: Grade 3 toxicity is closer to 34%, and forty-eight percent of patients required dose modifications. That 48% is the load-bearing number for the rest of this talk. For a drug being positioned as the backbone of future combinations, that's the headroom you start spending the moment you add a partner. The combination already moved to 200 mg — informally acknowledging the same headroom problem without searching for the actual optimum, which is the next slide. The question isn't "is daraxonrasib tolerable as monotherapy?" — clearly it is. The question is "at what dose, on what schedule, with what partners, in which patients?" That's a trial-design question, and the rest of this talk is about answering it.

Biomarker-driven combinations are the next move

- Daraxonrasib drives **RTK upregulation** on PDAC cell surface (Zhuang/Singh/Aronchik, Revolution Medicines, AACR RAS 2026)[3]
- Their framing: rational, biomarker-driven combinations to *pre-empt or overcome emergent resistance*[3]
- **1L combination cohort** (daraxonrasib **200 mg** + GnP, n=40, AACR 2026)[4]:
  - ORR: **58%** · DCR: **90%** · 6-mo PFS: **84%** · 6-mo OS: **90%**

Dose dropped from 300 mg to 200 mg for combination — but RMC-GI-102 fixed daraxonrasib at 200 mg without searching the joint surface. Informal OBD reasoning.

Zhuang et al., *Cancer Res* 2026 (AACR RAS) · Wolpin et al., AACR 2026 LB407



RTK upregulation is simultaneously a resistance mechanism and a rational combination target

Mallika Singh's group at Revolution Medicines has published the mechanistic case for what comes next. Inhibiting active RAS upregulates receptor tyrosine kinases on the cell surface — which is simultaneously a resistance mechanism and a rational combination target. Their explicit framing is that combinations should be biomarker-driven and resistance-preemptive. This is the manufacturer staking out the next decade of the program. And the first combination cohort has already delivered — 58% ORR with gemcitabine/nab-paclitaxel in first-line metastatic PDAC, 90% disease control, 84% six-month PFS. The combinations are coming. Many of them.

The dose-finding problem

Phase 1: *why MTD fails for RAS combinations, and what replaces it*

Why combinations break the rules you grew up on

- **3+3 was built for single-agent chemo**[5]. It assumes monotonic dose-toxicity, additive joint toxicity, Cycle-1 DLTs, and one-drug-at-a-time titration. **Combinations violate all four.**
- **In practice, combination dose-finding collapses to one drug.** Pick agent A's dose, fix it, add agent B. The joint efficacy-toxicity surface is **almost never searched** — in 162 combination trials, **88% used 3+3 and all but one used a single-agent design**[6,7].

Thall et al. (*Clinical Trials* 2024, methodologists' response to Project Optimus): conventional Phase I designs cannot reliably identify safe and effective doses[8].

Le Tourneau et al., *JNCI* 2009 · Rivière et al., *Ann Oncol* 2015 · Thall et al., *Clin Trials* 2024

3+3 answered the question it was built for — the MTD of a cytotoxic under a monotonic toxicity assumption. For RAS combinations none of those assumptions hold. The deeper problem is structural and operational. In practice almost no one searches the joint dose surface. The protocol fixes agent A's dose and adds agent B, so two monotherapy designs run in series and the joint surface is never characterized. Thall, Garrett-Mayer, Wages, Halabi, and Cheung made the formal version of this argument in their 2024 *Clinical Trials* paper, the SCT response to Project Optimus: conventional Phase I designs cannot reliably identify safe and effective doses. Q&A POCKET — frequentist alternatives. If a methodologist asks "what about non-Bayesian combination designs?" the honest answer is that very few practical, deployed frequentist combination designs exist. The field has converged on Bayesian / model-assisted designs (POCRM, BOIN-COMB, BLRM, PIPE, surface-free) because combinations require borrowing strength across dose pairs and partial-order constraints — naturally handled by priors and posterior updating, awkwardly handled without them. The cleanest frequentist alternative in the literature is Conaway, Dunbar, & Peddada 2004 (Biometrics 60:661-669), order-restricted isotonic regression for combinations — theoretically clean but not widely deployed. 3+3 generalizations to combinations exist but inherit all the original problems plus combinatorial explosion. Sequential lead-in (find A then add B) is the dominant industry practice — that is what we are criticizing. Q&A POCKET — Rivière backup. The Rivière et al 2015 systematic review of 162 combination phase I trials gives the "almost never" claim a documented count: 88% used 3+3, all but one used a design developed for single-agent evaluation. If anyone challenges "almost never" as an unsupported universal, that is your number. Next slide makes the abstract failure concrete with the Amgen sotorasib + pembrolizumab case.

The Amgen case: sotorasib + pembrolizumab

Same sponsor, two different designs.

- **Monotherapy (CodeBreaK 100):** Bayesian model-based design, escalation guided across the single-agent dose-response surface.
- **Combination (sotorasib + pembrolizumab):** IO dose fixed at standard, only sotorasib varied across cohorts. The joint surface was never modeled.

The defining toxicity proved the assumptions wrong twice:

- **Severe hepatotoxicity, not predictable from the single agent**
- **88% of Grade 3-4 events fell outside the Cycle 1 DLT window**[9] — a Cycle-1 design would have missed most of it.

Three of the four 3+3 assumptions failing at once: joint surface unsearched, toxicity non-additive, toxicity late-onset.

Hong et al., *NEJM* 2020 (CodeBreaK 100) · Li et al., *JTO* 2022 (WCLC OA03.06)

This is the worked example of the principle on the previous slide. The Amgen case is the clean one — and notice it is not a competence problem. Same sponsor used a model-based Bayesian logistic regression for sotorasib monotherapy in CodeBreaK 100. But for the sotorasib + pembrolizumab combination in CodeBreaK 100/101, they fixed the IO dose, explored only sotorasib across cohorts, and never modeled the two-drug surface. Even the most sophisticated sponsors collapse the combination to a one-drug search — which is the gap BATON closes. The combination's defining toxicity proved the point twice over: severe hepatotoxicity you could not predict from the single agent, and 88% of the Grade 3-4 events landed outside the 21-day DLT window — so a Cycle 1 design would have missed most of it. Q&A POCKET — where 88% comes from. The 22 of 25 numerator and denominator is the audit-ready figure. Source: Li et al WCLC 2022 OA03.06 / JTO 2022 supplement. If pressed, this single example demonstrates three of the four 3+3 assumptions failing at once: joint surface unsearched (combination not modeled), toxicity non-additive (synergistic hepatotoxicity), and late-onset toxicity outside Cycle 1 (88% out-of-window). The fourth assumption — monotonic dose-toxicity — wasn't directly tested here because they didn't search the surface.

Project Optimus: the regulatory ground has shifted

- FDA final guidance, 2024[10]: *Optimizing the Dosage of Human Prescription Drugs… for the Treatment of Oncologic Diseases*
- **Maximum Tolerated Dose → a dose optimized on benefit-risk (the OBD).**
- The expectation now is to **study more than one dose and justify the choice on efficacy and safety**, with **randomized dosage comparison** as the FDA's recommended approach.

This is no longer a methodological preference. It's the regulatory baseline.

FDA Guidance for Industry, 2024 (Project Optimus)

What you may not have caught is that the FDA has formalized this. The 2024 Project Optimus guidance moves oncology dose-finding from "maximum tolerated dose" to a dose optimized on benefit-risk — what the field calls the OBD. The guidance language is "optimized dosage," and the field's OBD shorthand maps to it directly. If you submit a Phase I protocol today that only characterizes the MTD, your reviewers are going to push back. This is no longer a methodological preference held by biostatisticians; it's the regulatory baseline. Now, on what the guidance actually expects: the FDA's stated default is to take two doses into Phase II — the MTD and a dose below it — and use a randomized comparison to determine which is better on benefit-risk. For a single agent that works. It does not scale to combinations, because you cannot randomize across a two-dimensional joint dose grid. That is exactly where model-based joint designs stop being a preference and start being the only practical way to meet the benefit-risk standard, which is where the rest of this section goes.

Why MTD fails for targeted agents and immunotherapies

- **Pembrolizumab** — efficacy plateaus at the lowest dose tested; MTD never reached[11]
- **CAR-T** — non-monotonic; high doses drive severe CRS/ICANS, not benefit
- **A RAS combination spans both shapes.** Neither rewards titrating to toxicity.

The reason MTD fails for these agents is mechanistic, and it has two flavors. I'll illustrate with pembrolizumab and CAR-T — non-RAS examples, but the cleanest in the literature, and the same dose-response shapes show up in the IO and bispecific partners daraxonrasib is going to be combined with. The first flavor you all know from pembrolizumab — the efficacy curve plateaus very early. In the original pembrolizumab studies, response rates across 2 mg/kg, 10 mg/kg every three weeks, and 10 mg/kg every two weeks were statistically indistinguishable. The exposure-response curve was flat. The MTD was never reached because efficacy never improved with dose. The 200 mg flat dose used today was selected by population PK modeling to reproduce the exposures seen at 2 mg/kg in a typical patient — a weight-based-to-flat conversion of the lowest active dose, not a search for the optimum. The second flavor is the umbrella. CAR-T is the canonical case. Very high cell doses unambiguously drive severe CRS and ICANS without proportional benefit. The downslope itself is product-specific — some show a true decline, others a plateau or threshold — but what is universal across CAR-T and bispecifics is the absence of a monotonic increase, and that alone is enough to break titrate-to-toxicity. These are not RAS drugs, but they are the cleanest pictures of the two shapes, and they are the shapes your RAS combinations will actually have. Daraxonrasib is the plateau — its monotherapy program never cleanly reached an MTD, had no DLTs, and 300 mg was selected on tolerability, which is exactly why 300 versus 200 is still a live question. The partners you'll add bring the umbrella. A RAS combination lives across both panels, and that is the whole reason MTD logic cannot find its optimal dose. You need a design that watches both dimensions.

The cost of MTD-only thinking

- **Sotorasib** (KRAS G12C, first-in-class, approved 2021)[12]:
  - 960 mg selected as the **highest dose tested**, no DLTs — a more-is-better default, not an MTD
  - FDA required a **240 mg vs 960 mg** comparison. **PFS was similar** at one-quarter the exposure[13]
- **MTD** **optimal dose** for targeted agents and immunotherapies
- **More than 40% of patients** in small-molecule registrational trials require dose reductions or interruptions[14]

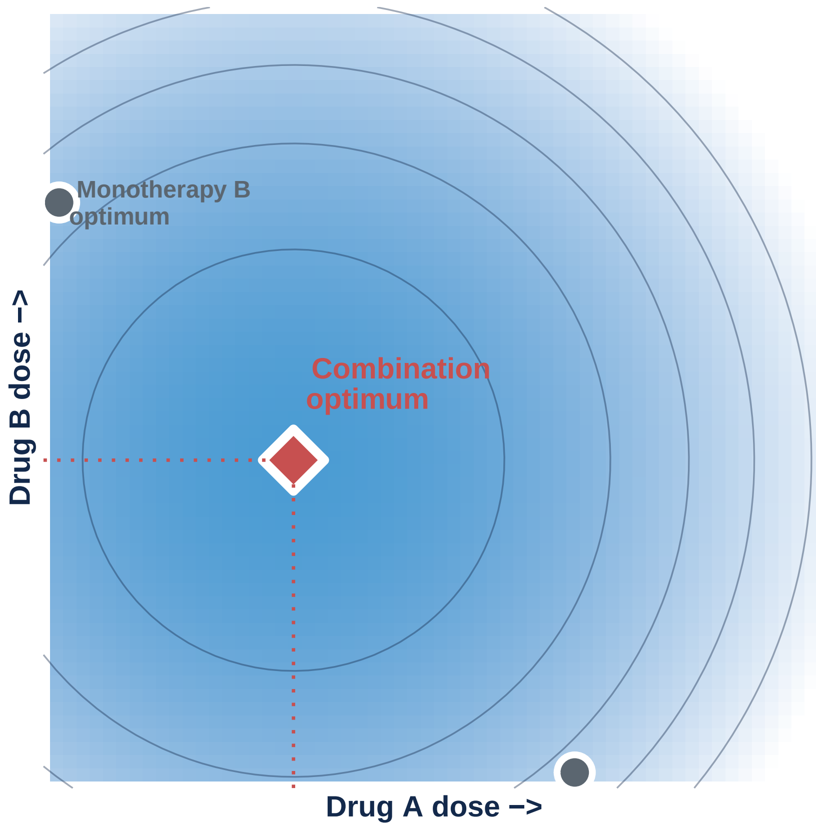
Same drug class as daraxonrasib.

Hong et al., *NEJM* 2020 (CodeBreaK 100) · Hochmair et al., *Eur J Cancer* 2024 · FDA–AACR, *Clin Cancer Res* 2025

The empirical evidence that this is not just theoretical. The FDA-AACR Strategies for Optimizing Dosages series, published in Clinical Cancer Research in 2025, reports that more than 40% of patients in small-molecule registrational trials require dose reductions or interruptions due to overly aggressive dosing — and links that pattern to potentially accelerated emergence of drug resistance, which quietly foreshadows the back half of this talk. The case you all know best is sotorasib, the first-in-class KRAS G12C inhibitor, approved in 2021. The Phase I program never reached an MTD because there were no DLTs; 960 mg was simply the highest dose tested, a more-is-better default rather than an MTD-driven choice. The FDA later required a post-marketing comparison against 240 mg, a fourfold lower dose. PFS at 240 mg was similar to PFS at 960 mg — the clean evidence that the approved dose was four times higher than needed on the primary endpoint. Note: I am deliberately not folding in the October 2023 ODAC vote on CodeBreaK 200 here. That 10-2 vote was about whether the confirmatory PFS read-out could be reliably interpreted given trial-conduct issues — informative censoring, imbalanced early dropout, investigator-assessed imaging — not about the dose. Conflating it with the dose story is the move a regulatory-savvy attendee will flag. The dose story alone is airtight; keep it that way. For daraxonrasib — same drug class — getting this wrong in Phase I means every combination on top of it inherits an unnecessarily toxic starting point.

Combination dose selection in RAS is harder

Combinations make MTD logic worse on every axis.



Schematic. The combination optimum sits below both single-agent doses; single-agent data alone cannot reconstruct the contour.

- The optimum can sit **below both single-agent doses**
- Single-agent data **cannot reconstruct** the joint surface
- For RAS, the **toxicity headroom is already spent** at the monotherapy dose

Fix what you know. Search what you don't.

Hong et al., *NEJM* 2020 · Wolpin et al., *NEJM* 2026 (RMC-6236-001)[2,12]

This is the bridge from the empirical cost — sotorasib at the MTD-only dose, daraxonrasib's already-spent headroom at 300 mg — into the methodological answer that follows. I am deliberately not re-establishing single-agent MTD failure here; the last three slides have already done that work. The job of this slide is to name what changes when you go from one drug to two. Three things to land verbally, supporting the one visible sentence. First, you are no longer choosing one dose, you are choosing a joint pair, and the single-agent margins do not reconstruct the surface — sotorasib's 960-versus-240 result tells you the surface looks different from what the single-dose escalation suggested. Second, effects interact on both efficacy and toxicity — the synergy point from assumption 4 — so the joint optimum can sit below either monotherapy dose. Third, and this is the RAS-specific part: the monotherapy toxicity is already substantial, 34% G 3 at 300 mg, 48% dose modification, so you start the combination with less headroom to spend. The first combination cohort already moved daraxonrasib down to 200 mg with GnP, but we do not know whether 200 mg is right or whether 150 mg or some other dose is right given other partners. The "fix what you know, search what you don't" callout is the operational rule: reduce to one drug only when the fixed component's dose is externally established — spend the search on the drug whose combination-optimal dose is genuinely uncertain. That is a dose-finding question that requires designs built for it. Which leads into the next slide.

Modern dose-finding: OBD instead of MTD

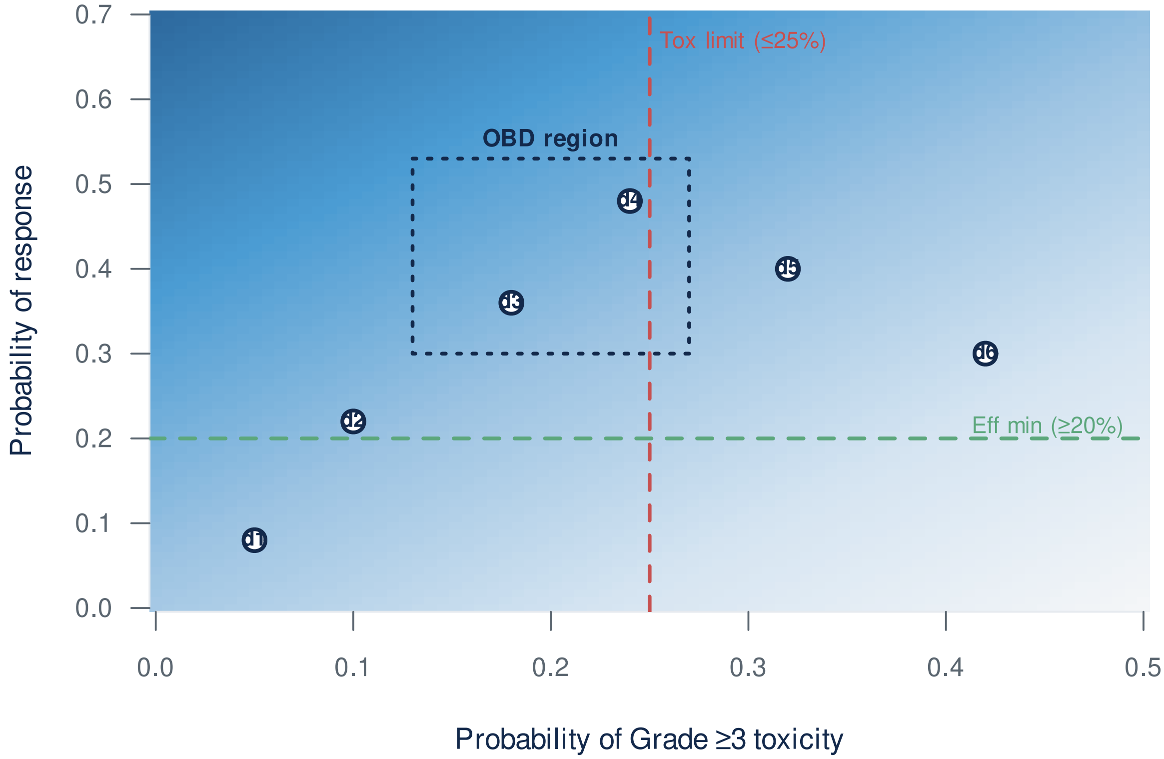
Phase I now ends with an OBD, not an MTD.

"A response is worth this; a Grade 3 toxicity costs this."

The team writes that utility down up front. The design — Bayesian, model-assisted — searches inside the safe envelope for the dose that **maximizes benefit-risk**, not toxicity.

Delivers the OBD Project Optimus calls for[10] — where randomized dose comparison can't scale to combinations.

Implementations: BOIN family[15,16] for printed-decision-table simplicity; POCRM, BLRM for combinations[17,18]. RAS Phase I already runs on these (sotorasib BLRM[12], adagrasib mTPI in KRYSTAL-1[19]).



Liu & Yuan 2015 · Zhou, Lee & Yuan 2019 (U-BOIN) · FDA Optimus Guidance

The paradigm shift in one sentence: Phase I now ends with an OBD, not an MTD. The how: before the trial starts, the team scores the trade-off — how do you weigh a response against a Grade 3 toxicity? That scoring becomes the navigation system. The design uses Bayesian / model-assisted rules to keep the trial inside a safe toxicity envelope while searching for the dose that maximizes the agreed-upon benefit-risk utility. What comes out is not a maximum tolerated dose — it's an OBD with a characterized benefit-risk profile. That's the dose Project Optimus is asking sponsors to deliver, and it's what you carry into Phase II combinations. ACRONYM POCKETS for Q&A — keep them OFF the visible slide. The BOIN family is the workhorse implementation: BOIN itself for single-agent (Liu and Yuan 2015), BOIN-COMB for drug combinations, TITE-BOIN for late-onset toxicity, BOIN12 for utility-based phase I/II, U-BOIN for the explicit two-stage utility design. POCRM (Wages and Conaway) and BLRM (Neuenschwander, Branson, Gsponer) are the established combination alternatives. mTPI is another common model-assisted variant; PIPE is a dual-agent product-of-independent-Beta-probabilities design (Mander 2015) for completeness. If a methodologist asks "which one?" the answer is: in the U.S. and academic groups, BOIN/U-BOIN dominate for the printed-decision-table operational simplicity; Novartis uses BLRM by default; the rest are specialty tools picked for specific constraints. Credibility line — leading RAS Phase I programs already run on Bayesian designs: sotorasib used BLRM in CodeBreaK 100, adagrasib used mTPI in KRYSTAL-1. The shift Project Optimus formalized is already industry standard for

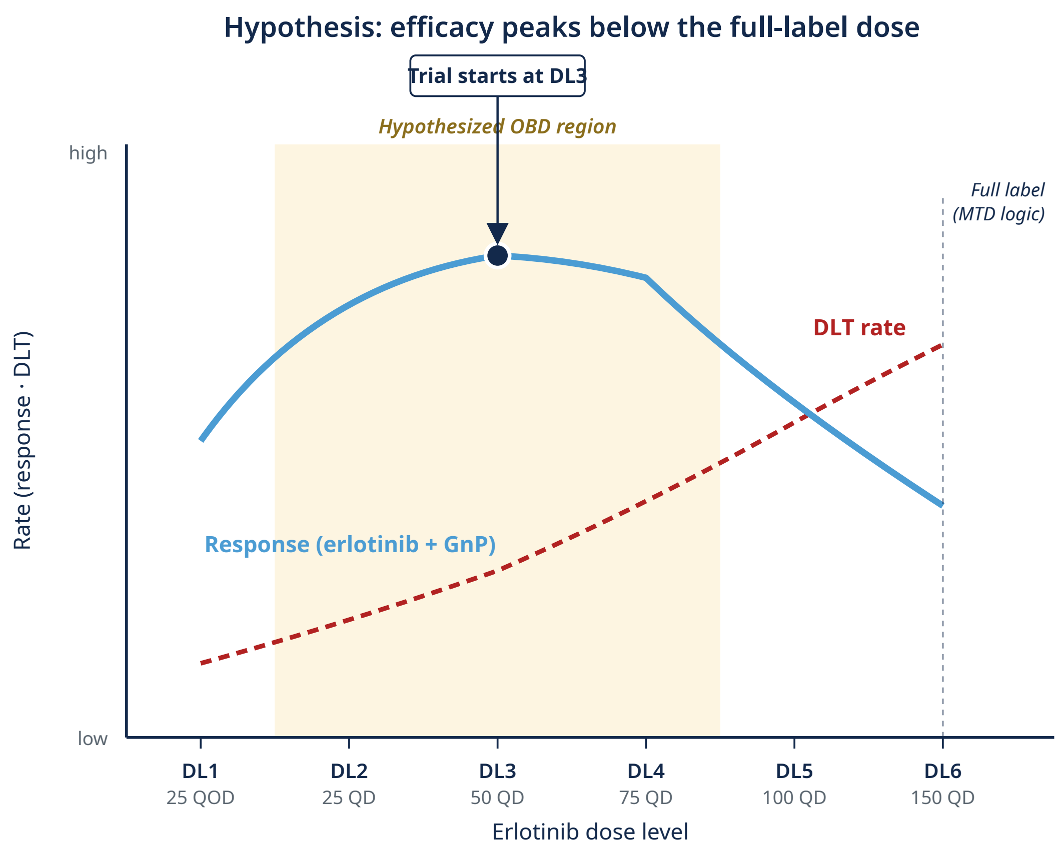
RAS monotherapy. For this audience the one-sentence takeaway is: modern Phase I optimizes benefit-risk instead of maximizing toxicity. Don't recite the acronym list out loud; only deploy it if a methodologist asks.

PANGEA: U-BOIN in basal-like PDAC

- **Trial:** LCCC2220-DCT (Somasundaram, PI), basal-like PDAC by **PurIST**[20]
- **Combination:** erlotinib + biweekly gemcitabine/nab-paclitaxel
- **Hypothesis:** lower-dose erlotinib may be more efficacious (OBD-below-MTD)[21]
- **Six dose levels:** 25 mg QOD → 25 / 50 / 75 / 100 / 150 mg QD; **start at DL3**

Design: *Stage 1 BOIN*[15] (target DLT 25%, cohorts of 4) → *Stage 2 BOIN12-like utility within U-BOIN*[16] (tox 25%, eff 20%). **Up to 52 patients; ~28 at OBD** (Scenario 1).

Rashid et al., *Clin Cancer Res* 2020 (PurIST) · Cohen et al., *Cancer Chemo Pharm* 2016 · Liu & Yuan 2015 · Zhou, Lee & Yuan 2019 (U-BOIN)

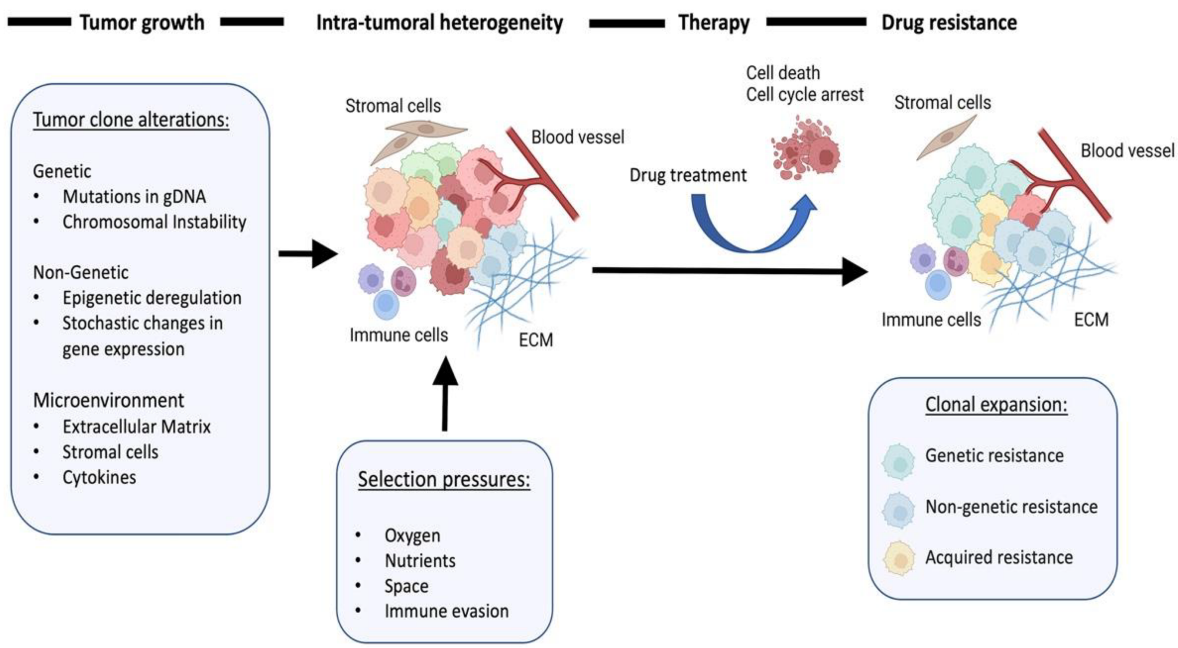


Here's the design in a real PDAC trial at UNC. PANGEA, led by Ashwin Somasundaram. The population is biomarker-defined — basal-like PDAC by PurIST, a subtype that has historically responded poorly to gemcitabine/nab-paclitaxel. The intervention is adding low-dose erlotinib to that backbone. And here's where the talk's argument lands on a real protocol: the central biological hypothesis of PANGEA is that lower doses of erlotinib may be more efficacious than the labeled dose, when given in combination with chemotherapy. The full-label dose of erlotinib is 150 mg daily. PANGEA tests six dose levels going down to 25 mg every other day. This is the OBD-below-MTD case made concrete — exactly the kind of question 3+3 cannot answer, and exactly the kind of question U-BOIN was built for. Stage 1 uses BOIN-style toxicity rules to establish the safe envelope, with a target DLT rate of 25%. Once we have enough data at any dose, we transition to Stage 2, which uses a utility function — elicited from our clinical team — to find the dose that maximizes the benefit-risk tradeoff. Doses have to clear both toxicity admissibility, no more than 25% DLT, and efficacy admissibility, at least 20% response rate. The trial ends when twenty patients have accumulated at any one dose, or when fifty-two patients have been treated total. Under the most plausible scenario from our simulations, we expect about thirty-eight patients in the U-BOIN phase, with about twenty-eight ultimately treated at the OBD. Q&A POCKET — justified dimensionality reduction. If a methodologist asks "why are you only searching erlotinib, not the joint surface?": GnP is fixed because the FDA-approved standard-of-care dose is established (MPACT); erlotinib is searched because its combination OBD-below-MTD is the genuinely open question. This is a justified single-axis search inside a two-drug regimen, not a regression to the fix-A-add-B antipattern. This is what an OBD-paradigm Phase I trial looks like in basal-like PDAC.

Resistance is the next problem

Phase 2 onward: *each patient is a moving target*

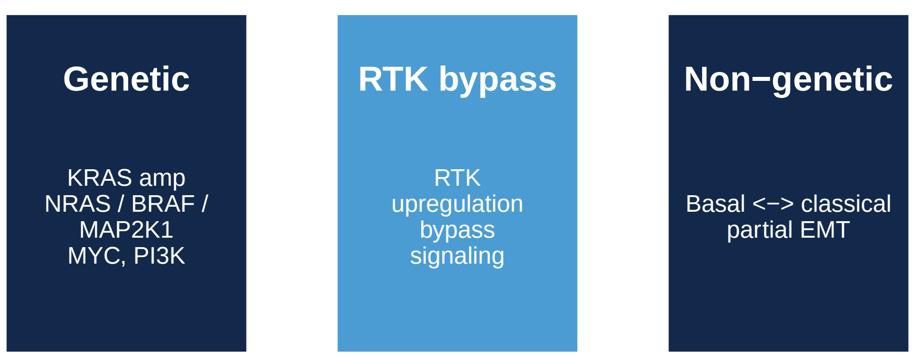
Resistance is a dynamic process



Schematic adapted from ARPA-H ADAPT program overview materials.

Resistance mechanisms emerge under RAS-inhibitor pressure

Daraxonrasib monotherapy in mPDAC (n=44, paired ctDNA): **59% show RAS-pathway resistance at progression — KRAS amplification in 36%**[22]

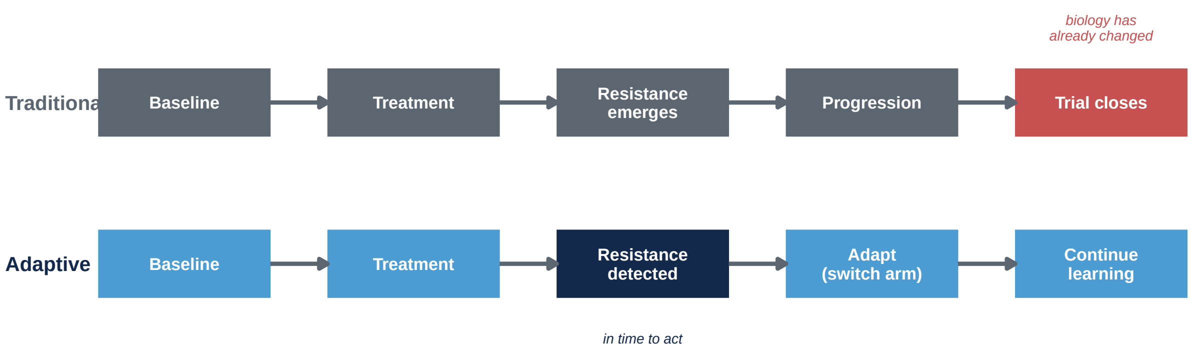


[3,23–26]

Classical trial design treats patients as if they're draws from a fixed distribution — enroll, randomize, administer the regimen, measure outcomes at predefined timepoints. That logic works when the underlying biology is static. Under sustained RAS-inhibitor pressure it doesn't.

The tumor evolves. Each patient is a moving target. Concretely — and this is the headline I want to anchor on — Aronchik's ENA 2025 paired-ctDNA work on the drug we just saw on slide 4, daraxonrasib, shows that 59% of PDAC patients on monotherapy develop detectable RAS-pathway resistance alterations at progression, with KRAS amplification specifically in 36%. Same drug from RASolute 302, same disease, n=44 patients with paired pre- and end-of-treatment ctDNA. That is the dynamic biology the trial design has to be prepared to see. Resistance signals start appearing weeks into therapy — sometimes earlier. The mechanisms split into a few buckets. Pathway reactivation through KRAS amplification or downstream NRAS, BRAF, MAP2K1 mutations — Aguirre's PDAC clinical work and Awad's adagrasib resistance series. Downstream amplification — MYC and PI3K family — is Wasko's preclinical RMC-7977 finding (the prevalent in-vivo resistance mechanism in KPC tumours, not pathway reactivation per se since pERK remained suppressed in resistant tumours). RTK upregulation and bypass signaling — the Zhuang and Aronchik mechanistic case for combinations. Cell-state plasticity between basal-like and classical with partial EMT — Aguirre again, plus the Yeh and Johnson labs' kinome work at UNC. None of these are baseline phenotypes. The biology is genuinely dynamic — and that's the setup for what trials need to do differently, which is the next slide.

Why classical trial designs miss resistance



Genomic → ctDNA · non-genomic / plasticity → biopsies, transcriptomics, imaging. A resistance-aware trial has to **do both** during enrollment[23,24].

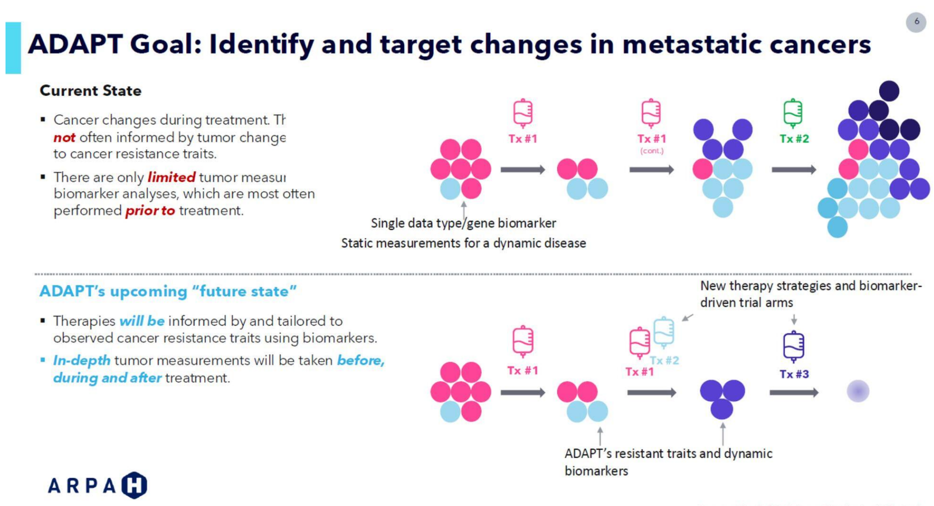
Awad et al., *NEJM* 2021 · Aguirre et al., *Cancer Discov* 2024

This is the design-implications side of the previous slide. The two-row timeline names the structural failure of classical trial designs under RAS-era biology. The traditional row: baseline → treatment → resistance emerges → progression → trial closes — and the red marker on "trial closes" is the point: by the time the trial reads out, the biology has already changed. The adaptive row catches resistance when it emerges, in time to act on it within the same protocol. Three things to land verbally. First, fixed regimens miss the interception window — by the time the trial reads out, the biology has changed. Second, baseline-only biomarkers can't track evolution under therapy — you measured the wrong thing at the wrong time. Third, resistance is characterized retrospectively, only after progression and trial closure. The Awad and Aguirre papers cited here are themselves post-progression analyses — ctDNA and biopsies collected after patients had already failed therapy. The field's best resistance knowledge was generated exactly the way I'm criticizing — after the fact, outside any prospective design built to catch it. Fourth, the temporal surface — when resistance emerges, in which patients, by which mechanism — is never characterized. The empirical anchor for this room: Aronchik's ENA 2025 paired-ctDNA work on daraxonrasib[22] shows RAS-pathway alterations — KRAS amplification (36% of patients), BRAF, MAP2K1 — emerging at progression in 59% of PDAC patients on monotherapy, mechanisms that a baseline-biomarker trial would not catch. The design implication splits with the biology: genomic mechanisms are catchable by ctDNA — cheap, continuous, easy to add. Non-genomic plasticity needs serial biopsies, transcriptomics, sometimes imaging — expensive, episodic, hard. A resistance-aware trial has to do both, during enrollment, not after. Same structural failure as the combination dose-finding critique two sections ago: static designs for a dynamic problem. Just as the dose problem needed Bayesian designs that handle the joint surface, the resistance problem needs designs that learn during enrollment. Next slide shows what those look like.

ADAPT / EVOLVE: what a resistance-aware trial looks like

Different disease, same design problem.

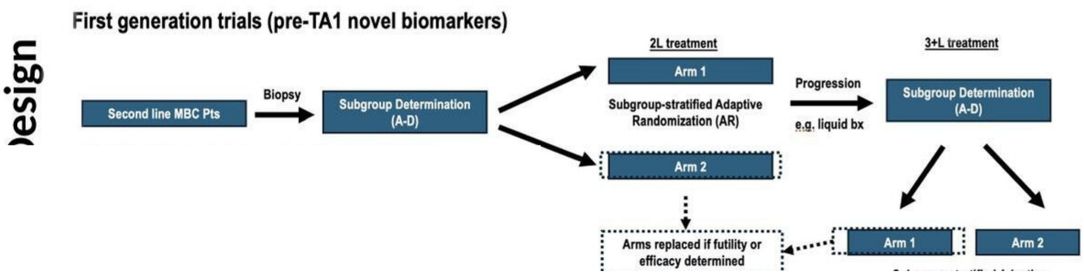
- Longitudinal resistance detection
- Biomarker-guided adaptation
- Template ports directly to PDAC (next slide)



ARPA-H **ADAPT** → **EVOLVE** (UNC-led; Carey PI, Rashid co-PI). Enrolling.

This is what a resistance-aware trial actually looks like operationally. ARPA-H — the new Advanced Research Projects Agency for Health — has funded a program called ADAPT specifically to build trials of this kind. The ARPA-H figure tells the story in two panels. Top: current state. A single static biomarker, sequential treatments — Tx 1, then Tx 1 continued, then Tx 2 — and the resistant subclones persist into the final cell mix. Bottom: ADAPT's future state. In-depth dynamic biomarkers, treatments switched in response to observed resistance traits, eventual clearance. Our EVOLVE trial in metastatic breast cancer is one of the first to operationalize this, and the design template generalizes directly to the kind of RAS-backbone combination trial you all are going to be running for PDAC in the next few years. CRITICAL FRAMING DISCIPLINE: the closer says "enrolling," not "EVOLVE shows" or "EVOLVE works." This is the resistance-side honesty I committed to in the opening — methods exist and are deployed, not yet completed. If a trialist asks for the readout, the answer is the design exists and is accruing; the readout is what we will report when interim looks are run. Do not let the live delivery drift toward an implied result. The next slide takes the audience inside the protocol — what EVOLVE actually does, and what the PDAC analog would look like.

EVOLVE-BDT — and what this would look like in PDAC



TBCRC EVOLVE-BDT flow — biomarker-stratified subtrials, adaptive randomization, in-trial arm replacement on futility or efficacy.

EVOLVE-BDT (real, enrolling)

ER+/HER2− and TNBC, biomarker-stratified subtrials with Bayesian arm-swap on futility/efficacy.

PDAC analog (hypothetical)

Basal-like vs classical by PurIST[20], same Bayesian arm-swap logic on daraxonrasib-combination arms.

Same architecture, different disease.

This is the load-bearing slide for "could we actually do this in PDAC?" The figure is EVOLVE-BDT's flow — TBCRC EVOLVE-BDT, our group's adaptive platform in 2L metastatic breast cancer. Walk it left to right. Second line MBC patient enters. Biopsy. Subgroup determination — molecular profile assigns A through D. Subgroup-stratified adaptive randomization sends the patient to 2L Arm 1 or Arm 2. Patient is treated. At progression, a liquid biopsy re-determines the subgroup — because the biology has changed — and the patient enters 3+L treatment in a subgroup-stratified Arm 1 or Arm 2. Crucially, the dashed boxes mean arms get replaced when the Bayesian decision rule fires futility or efficacy. New arms enter. The trial keeps learning. Now the columns below the figure are the explicit translation. Left: real EVOLVE specs. ER+/HER2- and TNBC parallel platforms, about 700 patients total. Subgroups A through D defined by ESR1 mutations, PI3K/AKT/PTEN status, PD-L1, androgen receptor — all things that already drive treatment decisions in breast clinical practice. Serial tumor and blood at every visit. Arms swap on the Bayesian decision rule. Right: the PDAC translation. Stratify by basal-like versus classical using PurIST. Subgroups by KRAS amplification, RTK upregulation, partial EMT — the Aronchik data shows 59% develop measurable resistance, so these subgroups have empirical basis. Serial ctDNA for the genomic side, serial biopsies for plasticity — the do-both requirement made concrete. Arms include daraxonrasib plus EGFR-inhibitor for RTK-upregulation patients, daraxonrasib plus SHP2-inhibitor for pathway-reactivation patients, classical-state shifters for plasticity-driven failures. The bottom callout makes the point: same architecture. Every row of the right column has a real biomarker, a real assay, and a real combination partner already in or entering the clinic. This is not a hypothetical that needs new science. It's the deployment problem. Which is exactly what BATON solves. Pause briefly before moving on to the statistical-demands slide.

What this demands of the statistical design

Four design axes — each disease-agnostic. A RAS-backbone PDAC combination would need all four for the same reasons.

Axis	Operationalized as
Adaptive stopping	Bayesian futility/efficacy boundaries at each interim
Real-time biomarker discovery	Pre-specified candidate signatures + agnostic discovery window
Longitudinal monitoring	Serial ctDNA, imaging, biopsies on every patient — *the "do both" requirement*
In-trial evaluation	Biomarker-guided cohorts within the same protocol — *characterize the temporal surface and act on it*

Continual learning during enrollment is not a tweak to a Phase II — it is a

structurally different trial. Each axis adds interacting parameters: interim timing, posterior-probability thresholds, biomarker decision rules, randomization-weight updates, cohort-opening logic.

The methods exist.**Calibrating them to actually launch is the gap.**

Four axes that operationalize the resistance-aware trial template. I am showing them to GI SPORE leaders because they are disease-agnostic — requirements for any resistance-aware trial, not properties of breast biology. A RAS-backbone PDAC combination would need all four for the same reasons. Adaptive stopping rules — Bayesian futility and efficacy boundaries that update at every interim look. Real-time biomarker discovery — pre-specified candidate signatures plus an agnostic discovery window so you're not locked into your initial hypothesis. Longitudinal monitoring — serial ctDNA, imaging, biopsies on every patient. This is the "do both" requirement from the mechanisms slide made concrete: ctDNA for the genomic side, biopsies and transcriptomics for the non-genomic plasticity side, on every patient throughout the trial. And in-trial evaluation — biomarker-guided cohorts opened within the same protocol, so when you find a resistance signature you can test the strategy that addresses it during the trial, not as a follow-on study three years later. Now the load-bearing point that sets up the rest of the talk. Continual learning during enrollment is not a tweak to a Phase II — it is a structurally different kind of trial. Each axis adds interacting parameters to the design: interim look timing, posterior-probability thresholds for stopping, biomarker decision rules, randomization-weight updating, cohort-opening logic. Bayesian adaptive designs that handle all of this exist — posterior-probability monitoring, simulation-based operating characteristics, the whole toolkit. What does NOT yet exist routinely is the infrastructure to make those designs deployable — to calibrate them against the operating-characteristic targets that the trial team and the FDA agree on. That's the gap. The next section is about closing it.

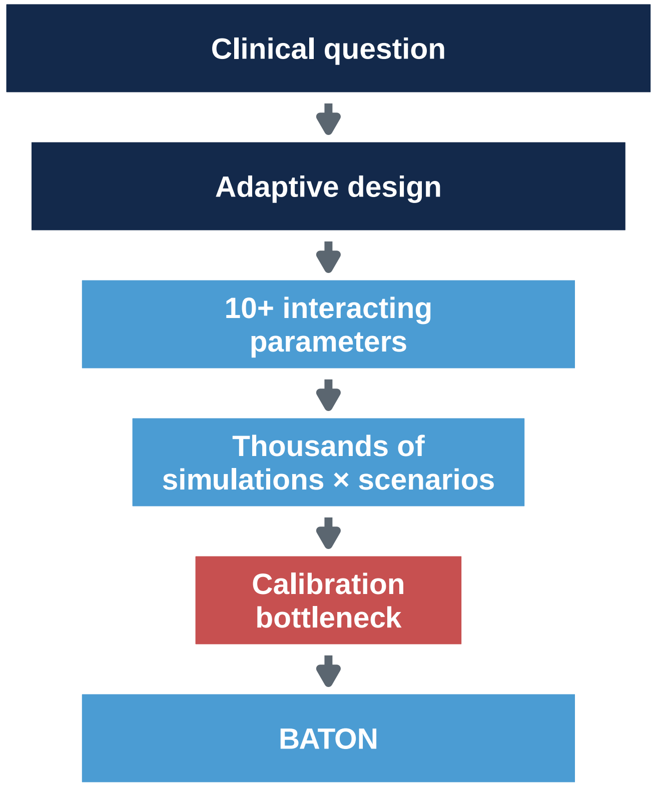
Putting these into practice

Why don't we see more of these trials?

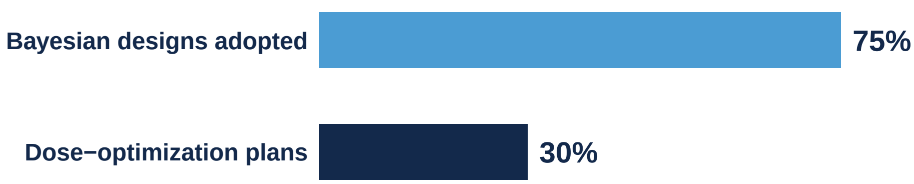
Why don't we see more of these trials?

Funding, accrual, and biopsy logistics are real.

But the barrier that quietly kills good designs — and the one we can actually remove — is calibration.



- Operating characteristics (Type I, power, N, duration) depend on every parameter, **in interaction**
- Design space: **8–13 dimensions** for trials we've calibrated; **20+** for combined dose + resistance. Grid search is infeasible.



Methods adopted. Dose-optimization plans — where calibration cost bites — not[27].

Our group: several months per design. For combined dose + resistance designs, hand-tuning is infeasible.

FDA–AACR Project Optimus impact review, *JCO Onc Adv* 2025 (367 Phase I protocols)

This is the slide that opens the BATON section, and I want to land it carefully. Open with the concession explicitly: funding, accrual, biopsy logistics — these are real, and this room has lived them. I am not standing up here telling you the barrier to better trials is statistics, because that would be a biostatistician naming his own contribution as the field's central obstacle, and you would be right to push back. What I am claiming is narrower and more specific: among the barriers, calibration is the one that quietly kills good designs and the one we can actually remove. You have all watched PIs default back to 3+3 even when they knew a better design existed — the calibration cost was the hidden reason, and that is the gap BATON closes. Now the body. The methods I just walked through — modern dose-finding for combinations, longitudinal biomarker-aware adaptive designs — exist. The reason every clinical trial isn't using them is calibration cost. Every modern design has knobs: target DLT and efficacy rates, utility weights, cohort sizes, stopping rules, sample-size caps. Operating characteristics depend on every knob, in interaction. To know whether a design meets Type I, power, expected N, expected duration, OBD-selection accuracy, and biomarker-promotion error control, you simulate thousands of trial replicates per candidate. And you do this across plausible scenarios. The design space is eight to thirteen dimensions for the trials we've calibrated, and twenty-plus when you combine dose-finding with resistance tracking. A grid search is hopeless — both computationally infeasible and statistically wasteful, because most of the parameter space gives bad designs. In my group, calibrating a single modern combination Phase I/II design by hand takes a senior statistician several months of focused work. For trials that handle dose-finding AND resistance tracking simultaneously, you're at twenty-plus parameters with even more constraints. Hand-tuning doesn't just get slower; it gets infeasible. That sets up the pivot — the next slide.

"We could design the science.

We could not design the trial."

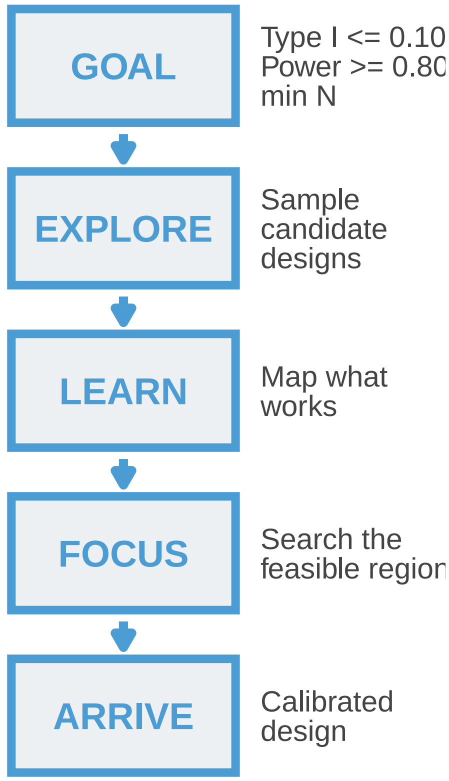
Pause. Let the line land. This is the pivot of the talk. Everything before this slide has been about establishing that the methods exist — combination dose-finding, OBD selection, resistance-aware longitudinal designs. ADAPT and EVOLVE prove these methods are deployable. But during EVOLVE's activation, the team kept running into the same wall: we knew what trial we wanted to run, we knew the statistical machinery existed, and we still could not get a calibrated, FDA-defensible design out the door. The methods existed. Deploying them did not. That is the gap. Pause briefly. Then move on to what closes it: BATON.

BATON: the calibration tool

BATON = **B**ayesian **A**daptive **T**rial **O**ptimizatio**N**[28]

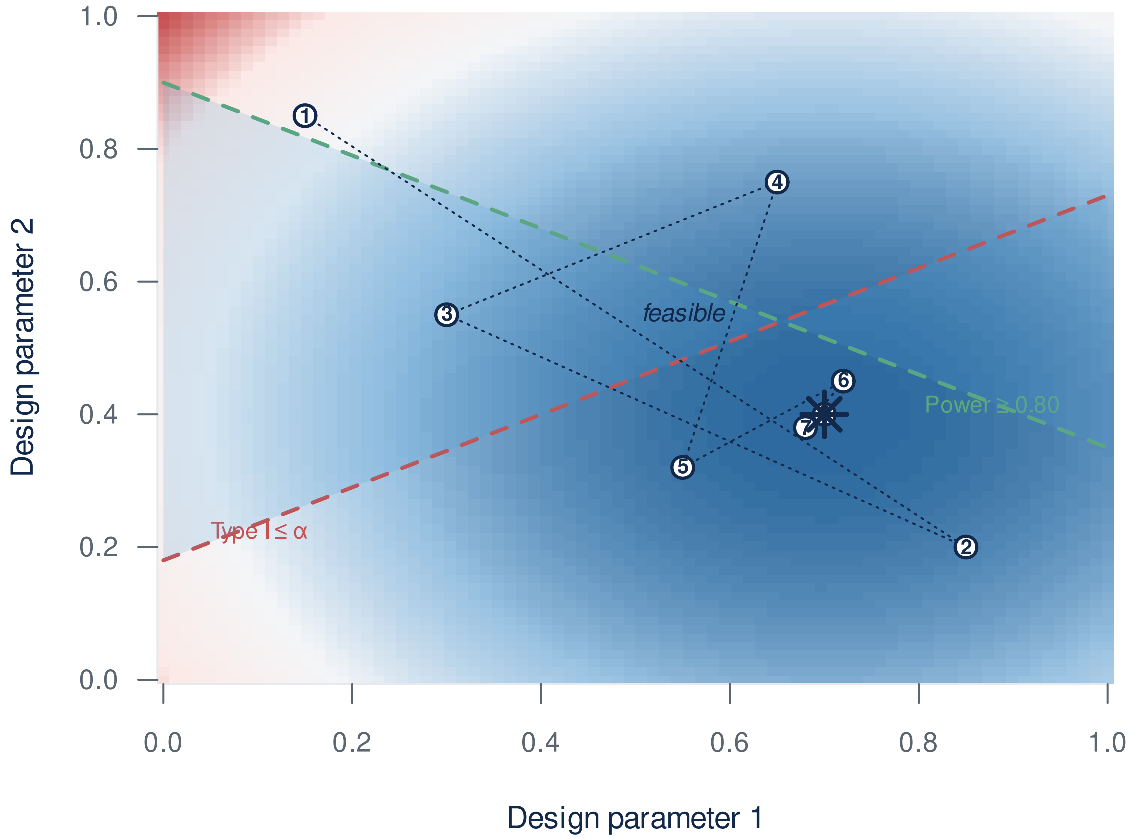
BATON turns clinically necessary trial complexity into calibrated, launchable designs.

Think of BATON as a navigation system for trial design:



Designs that would have taken months to discover — or never been found at all.

You bring the clinical design; BATON evaluates hundreds of candidate parameter settings *in silico* and returns the ones that meet your targets. R package: github.com/naimurashid/BATON

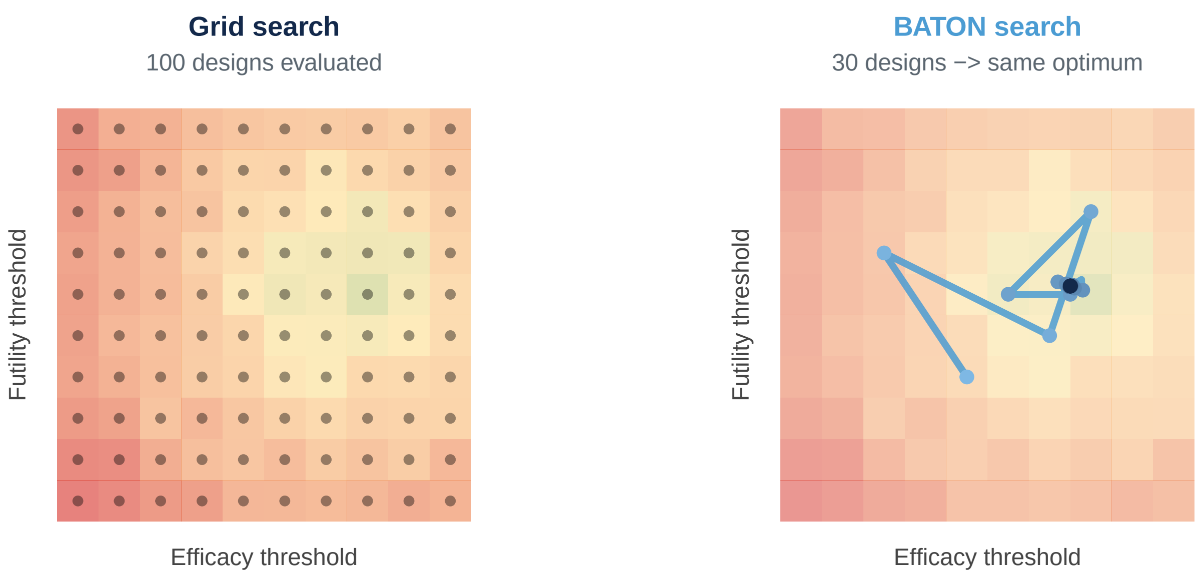


Red = high expected N (bad); blue = low. Dashed lines = Type I and Power constraints. Numbered points = BATON's sequential evaluations converging to the constrained optimum ( ).

Young et al., 2026 (BATON manuscript) · github.com/naimurashid/BATON

Think of BATON as GPS for trial design. You tell it the destination — power, type-I error, sample-size targets, expected duration. It explores candidate designs by Monte Carlo simulation. It learns which regions of the parameter space satisfy your constraints. It focuses the search on the feasible region. And it arrives at a calibrated design. Hours instead of months. Under the hood: constrained Bayesian optimization with Gaussian process surrogates, but for this audience what matters is the input and the output. You give BATON a design template, a set of operating-characteristic targets, and a scenario grid. BATON returns a calibrated design that meets your targets. FRAMING DISCIPLINE — and this matters because the next slide is PANGEA: in the BATON manuscript, the worked example is EVOLVE, not PANGEA. BATON was developed and validated during EVOLVE activation, across breast cancer subtrials at 8 to 13 parameters. PANGEA is an application of the same tool, not what the paper reports. If anyone asks "is this published," the clean answer is: the method and its validation are under review at JASA Applications and Case Studies; PANGEA is one application of the same software. Q&A POCKETS, in order of likelihood. (1) "Is BATON really optimal?" Manuscript benchmarked it against Simon two-stage designs where the optimum is known analytically; BATON converges to within 1% of Simon-optimal expected N, hitting the optimum exactly in 8 of 10 seeded runs. So the answer is: it recovers the analytically optimal design on problems where we can check. (2) "Can we use it?" Yes — open-source R package at github.com/naimurashid/BATON, with the companion evolveTrial package. (3) "What's the gain over grid or random search?" The paper has a head-to-head: BATON found feasible designs in 150–200 evaluations where grid and random search often found none at all. (4) "Why isn't a good statistician enough?" Honest answer: BATON IS the statistician's tool. A senior statistician with BATON > the same statistician without it. What BATON removes is the months-of-recalibration tax that currently makes the right designs operationally infeasible — the team can't commit to deliver a calibrated, defensible design within the trial timeline. Manual calibration also returns one design and you hope it's robust; BATON returns the feasible region and lets you pick the most defensible point in it. Under the hood for the methodologist who asks: constrained Bayesian optimization with Gaussian process surrogates over the operating-characteristic landscape.

Grid search vs BATON



70% fewer evaluations — minutes instead of days.

Color: design quality at each parameter combination (green = meets OC targets, red = does not). Path: BATON's sequential evaluations converging on the feasible optimum.

This is what's happening underneath the GPS metaphor. On the left: a grid search — 100 designs evaluated on a 10-by-10 lattice across two parameters. Most are nowhere near the feasible region. On the right: BATON. Same parameter space, same quality landscape underneath, but BATON only evaluates 30 designs — and it converges on the same optimum because each design choice is informed by what the previous evaluations revealed. Seventy percent fewer evaluations. Each evaluation is thousands of Monte Carlo trial replicates, so saving 70 percent of the search is the difference between hours and weeks. This is a 2D illustration; PANGEA Phase II's design space is 4-dimensional and the EVOLVE designs go up to 8 or 13 dimensions, where the search-efficiency advantage compounds rapidly.

Traditional calibration vs BATON

Manual approach

Aspect	Limitation
Process	Trial-and-error
Duration	Weeks to months
Scope	One design at a time
Reproducibility	Statistician-dependent
Documentation	Often incomplete
Regulatory risk	Variable

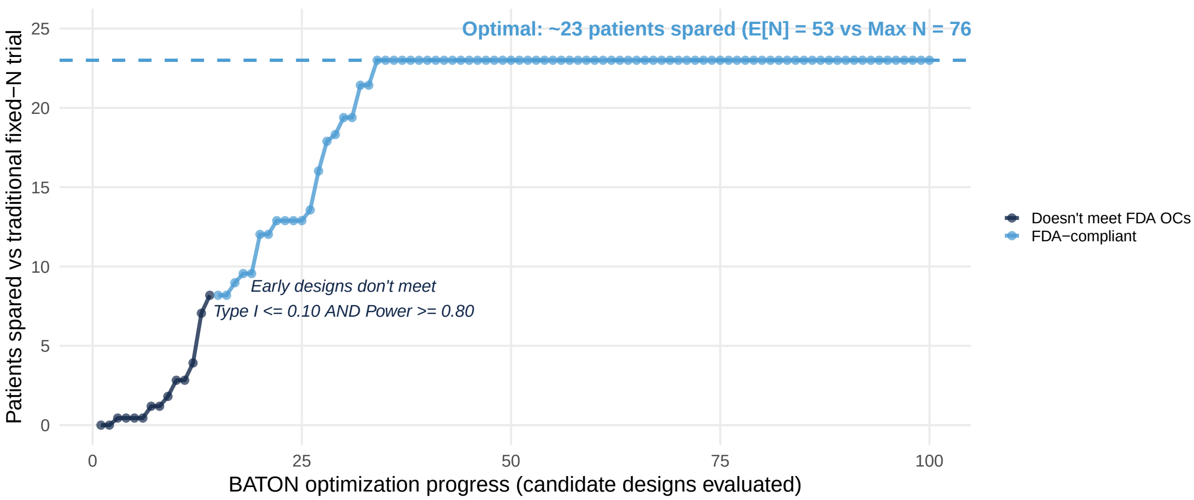
BATON

Aspect	Capability
Process	Systematic search
Duration	Hours wall-clock
Scope	Hundreds evaluated
Reproducibility	Deterministic (seeded)
Documentation	Complete FDA audit trail
Regulatory risk	Documented

Same clinical question. Same statistical constraints. The bottleneck wasn't the design — it was the calibration.

Same clinical question. Same statistical constraints. The bottleneck was never the design itself — it was the calibration. Hand-tuning is trial-and-error: takes days to weeks of a senior statistician's time, varies by who's doing it, produces incomplete documentation, leaves regulatory risk on the table. BATON systematically searches hundreds of designs in hours, with deterministic seeding and a complete FDA audit trail. The methodology shift Project Optimus formalized — adaptive Bayesian designs as the standard — is the destination. BATON is what makes getting there feasible at scale.

BATON's output, framed as patients spared



Bottom line: BATON-calibrated adaptive design with **E[N] = 53 vs Max N = 76**: ~23 fewer patients on a potentially-ineffective arm — while meeting Type I 0.10 AND Power 0.80. *The savings come from adaptive stopping; BATON's job is finding the design that achieves them within the OC constraints.*

Each point = one candidate design evaluated (10,000 simulated trials per design).

This is the same BATON convergence story you just saw on the grid-vs-BATON slide, but framed in clinically relevant units. Y-axis: patients spared relative to running a fixed-N=76 design. X-axis: BATON's optimization progress — each point is one candidate design that BATON evaluated by simulating ten thousand trials at that parameter combination. Dark-blue points are designs that don't yet meet the FDA operating-characteristic constraints — Type I error 0.10 AND power 0.80, simultaneously. Light-blue points are designs that do. BATON enters the feasible region around evaluation 14, then climbs to the optimum near evaluation 75. The horizontal dashed line — about 23 patients spared — is where the optimal calibrated design lands. CREDIT DISCIPLINE: the patients are spared by the adaptive design class itself — early futility/efficacy stopping is what makes E[N] of 53 possible against a max of 76. BATON's contribution is finding the configuration of stopping thresholds and futility margin that delivers those operating characteristics. Say "the BATON-calibrated adaptive design spares ~23 patients" or "the design BATON found spares ~23 patients", not "BATON spares ~23 patients" — the second sentence credits the optimizer rather than the design. The achieved expected N of 53 against the calibrated max of 76 is what you'll see on the next slide.

PANGEA Phase II: calibrated by BATON

- After Phase I identifies the OBD, PANGEA continues into a **Phase II Bayesian adaptive randomized trial**: GnP + erlotinib (at OBD) vs GnP alone
- Primary endpoint: PFS. **Design assumes** reference median 5.5 mo[29]; powered to detect 9.2 mo under H (HR 0.58)
- **BATON-calibrated** four parameters (N, efficacy threshold, futility threshold, futility margin)[28]
- **Constraints met:** Type I   0.10, Power   0.80

Achieved operating characteristics:

Metric	Value
Power	**82.2%**
Type I error	7.0%
Max N	**76** (82 accrual w/ buffer)
Expected N under H	53.0
Expected N under H	58.1
Pr(early futility stop | H)	**77.9%**
Pr(early efficacy stop | H)	**81.4%**
Expected duration	27.3 mo

Von Hoff et al., *NEJM* 2013 (MPACT) · Young et al., 2026 (BATON manuscript)

Once U-BOIN selects the OBD, PANGEA transitions into a Phase II Bayesian adaptive randomized trial — basal-like PDAC patients randomized one-to-one to GnP plus erlotinib at the OBD versus GnP alone, with PFS as the primary endpoint. TENSE DISCIPLINE — this is critical and the slide structure protects it but the live delivery needs to as well. The 5.5-month reference, the 9.2-month alternative, and the HR 0.58 are DESIGN ASSUMPTIONS, not observed results. The 5.5 is the GnP doublet benchmark from MPACT, Von Hoff 2013. The trial is powered to detect a hazard ratio of 0.58, not showing one. The operating characteristics in the right column ARE achieved, because they are the outputs of BATON's calibration — Type I at 7.0%, power at 82.2%, those are properties of the design that BATON found, computed over 10,000 simulated trials per scenario. So: design assumptions on the left, calibrated achievements on the right. Never let "BATON found a design that achieves these OCs" drift to "the trial showed this." The design uses a Bayesian piecewise exponential survival model with interim analyses for early stopping on futility or efficacy. Four design parameters had to be set jointly to meet operating characteristics — the maximum sample size, the efficacy stopping threshold, the futility stopping threshold, and the futility hazard-ratio margin. CONFIRM AGAINST MANUSCRIPT before delivery which exact four BATON searched versus which were fixed by clinical/modeling choice, since a methodologist may ask. BATON found a configuration that meets Type I error at most 0.10 and power at least 0.80, simultaneously. The achieved characteristics are on the slide. Three numbers I want to draw your attention to. First, the power is eighty-two percent at a Type I error of seven percent — solidly within the targets. Second — and this is the clinical payoff — the trial stops early for futility seventy-eight percent of the time when the drug doesn't work. Patients are not being randomized to a losing arm waiting for the readout. Third, the trial stops early for efficacy eighty-one percent of the time when the drug does work. That's how quickly modern adaptive designs can deliver an answer when there's signal. Expected trial duration is twenty-seven months, with expected enrollment of fifty-three patients under the null and fifty-eight under the alternative — well below the calibrated max of seventy-six. (Note: the protocol narrative carries 82 as the accrual target with buffer; 76 is the calibrated design maximum. If anyone asks, "76 is the calibrated maximum, 82 is the accrual goal." Action item for me separately: reconcile the protocol so a reader does not catch the discrepancy.) This entire calibration was done by BATON. Hand-tuning a design with these four interacting parameters against two simultaneous operating-characteristic constraints, across thousands of simulated trials, is the work I was describing earlier as several months of senior statistician time. BATON delivered this configuration in hours.

What BATON changed for PANGEA Phase II

Without BATON

- Months of senior-statistician hand-tuning
- Four interacting parameters; designs typically fail Type I 0.10 **and** Power 0.80 simultaneously
- No reproducibility guarantee; no audit trail
- Activation slips, *or* design ships under-validated

With BATON

- **150–200 evaluations**, hours wall-clock
- All four parameters jointly calibrated against both constraints
- Deterministic, FDA-grade audit trail
- Achieved OCs from the previous slide — power 82.2%, type I 7.0%, expected duration 27.3 mo

PANGEA Phase II is the proof: a calibrated, FDA-defensible adaptive design — delivered in days, not months.

Young et al., 2026 (BATON)

This is the practical-impact slide. Without BATON: months of a senior statistician's time hand-tuning, designs that typically fail to hit Type I 0.10 AND Power 0.80 simultaneously across the four interacting parameters, no reproducibility guarantee, no audit trail — so either activation slips or the design ships under-validated. With BATON: 150 to 200 evaluations, hours of wall-clock time, all four parameters jointly calibrated against both constraints, deterministic seeded output, FDA-grade audit trail. The achieved operating characteristics on the previous slide — power 82.2%, type I 7.0%, expected duration 27.3 months — those came out of BATON, not out of months of manual iteration. Delivered in days, not months. The argument is not "the statistics got fancier"; the argument is "the bottleneck moved." Calibration used to be the choke point. BATON moves it.

What this means for a PDAC patient

Traditional fixed Phase II

Mr. Hernandez, 62, mPDAC, basal-like by PurIST, progressed on 1L GnP

- Enrolls in standard randomized Phase II of GnP + erlotinib vs GnP alone
- Trial enrolls all **76 patients** before final analysis
- Three-plus years to read out
- Drug effect (if any) discovered only at study end

PANGEA Phase II (BATON-calibrated)

Mr. Hernandez, 62, mPDAC, basal-like by PurIST, progressed on 1L GnP

- Same randomization; interim analyses on accumulating PFS events
- **78% chance** of early futility stop if erlotinib adds nothing
- Expected enrollment: ~**53 patients** (H) / ~**58** (H)
- Expected duration: **27.3 months**

Bottom line: ~23 fewer patients on potentially-ineffective therapy. Answer in 27.3 months instead of 36+.

This is what calibrated PANGEA Phase II means for a real patient. Pick someone in your clinic. A 62-year-old with metastatic PDAC, basal-like by PurIST, progressed on first-line gemcitabine plus nab-paclitaxel. In a traditional fixed Phase II of GnP plus erlotinib versus GnP alone, that patient enrolls into a trial that will run all 76 patients before anyone learns whether erlotinib adds anything. Three-plus years to read out. In PANGEA Phase II, calibrated by BATON: same randomization, but interim analyses run as PFS events accumulate. The design has a seventy-eight percent chance of stopping early for futility if erlotinib adds nothing — patients are not randomized to an inactive arm beyond what's needed to learn that. Expected enrollment is around 53 patients under H0 and 58 under H1. Expected duration: 27.3 months, not 36-plus. About 23 fewer patients exposed to a potentially-ineffective therapy. That is the clinical payoff of calibrating with BATON instead of shipping a fixed-N design that learns slowly. This is one trial. When the calibration becomes routine for the GI portfolio, the savings compound across every adaptive trial we activate.

An unexpected finding

Smaller trials are not always better.

Over-constraining the max N can *prevent* patients from learning sooner that the treatment works.

BATON surfaced this during PANGEA Phase II calibration: caps set too low suppress early-efficacy declarations. Manual calibration anchors on the first feasible design; BATON returned the Pareto frontier.

A discovery story from PANGEA's BATON calibration. We started this design with an instinct toward a tight max N — strict capacity, minimize regret if the drug doesn't work. BATON identified a non-obvious trade-off: pushing the cap too aggressively didn't just constrain the worst case, it actively suppressed the early-efficacy declarations. Caps set too low force the trial to wait for the final analysis before being able to claim efficacy. Patients wait longer for an answer when the drug DOES work. This was not a tuning preference; it was a structural property of the design space that BATON surfaced because it explored the full feasible region, not just one corner of it. The recommendation was a slightly higher cap that preserves the futility-stopping rate while also preserving early-efficacy declarations. This is the kind of insight manual calibration misses because manual calibration anchors on the first feasible design it finds. BATON returns the Pareto frontier; you choose where on it to land. The 76 you saw on the PANGEA OCs slide is that choice.

Where this is going

Three design philosophies, in plain language. BATON calibrates each:

- **Aggressive** — minimize expected N; find active drugs fastest *(statistical name: Optimal / Fleming)*
- **Conservative** — bounded max N; strict capacity *(Minimax)*
- **Balanced** — Pareto-optimal compromise between the two *(Admissible)*

Every point on the Pareto frontier is a valid design. The choice is about what your program needs most.

Where this lands in GI:

PDAC

RAS−backbone combinations
U−BOIN / BOIN12

CRC

Joint−surface novel−novel doublets
POCRM · BOIN−COMB · BLRM

Basket / umbrella

Biomarker−stratified subtrials
shared decision rules

Robust calibration

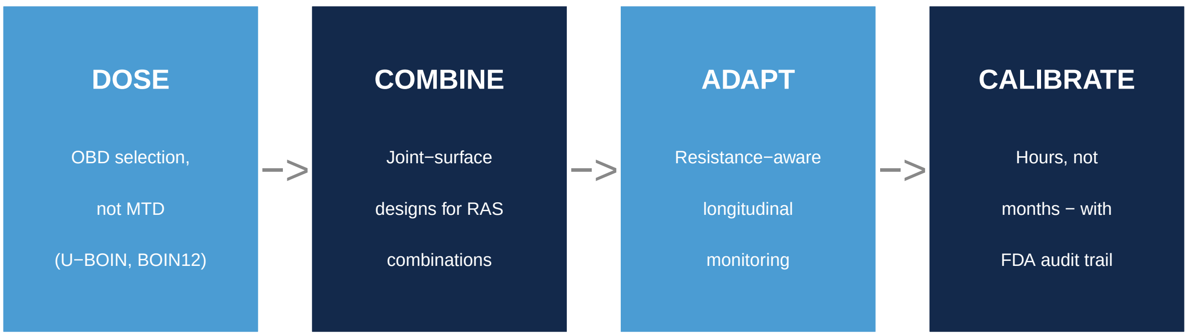
CARBO − designs robust to uncertain assumptions

Could BATON become standard infrastructure for adaptive trial design in GI oncology?

Young et al., 2026 (BATON) · Wages 2011 (POCRM) · Lin & Yin 2017 (BOIN-COMB) · Neuenschwander 2008 (BLRM)

Q&A POCKET — non-RAS examples of joint-surface combination dose-finding. If anyone asks “do any trials actually use joint-surface combination designs”: yes, two clean concrete answers. (1) POCRM: the Embolden trial (NCT03240211) at UVA — pembrolizumab + pralatrexate/decitabine combination Phase Ib in T cell lymphomas, published implementation by Mozgunov et al, Stat Med 2022. (2) BLRM: Novartis Oncology has used BLRM as its standard Phase I design for over a decade and now exclusively for Phase 1 — including combination trials via the dual-agent extension. So joint-surface designs are not theoretical — they have published RCT deployments and one of the largest oncology developers in the world uses BLRM by default. The RAS-specific gap is that novel-novel RAS doublets like elironrasib + daraxonrasib are in the clinic but have not yet adopted these methods. That gap is exactly where BATON’s joint-design extension lands. Where this is going, in two pieces. First, what’s done. BATON has calibrated both EVOLVE — the single-arm, between-arm, and seamless two-stage breast cancer subtrials reported in the BATON manuscript — and PANGEA Phase II, the calibration you just saw. Two trials, two diseases, same tool. Second, the extensions in flight. The methodological move that matters most for this audience is bringing BATON’s constrained-Bayesian-optimization machinery to the Phase I family — U-BOIN and BOIN12 — for OBD selection. PANGEA Phase I uses U-BOIN today, calibrated with the standard online software; the next step is doing that calibration under BATON so the multi-parameter Phase I design space stops being the bottleneck. That’s how the calibration cost that has historically pushed PIs back toward 3+3 disappears. The second extension is CARBO — Context-Aware Robust Bayesian Optimization — which calibrates against a range of plausible trial assumptions rather than a single point estimate. This matters when the assumption was X but the trial actually enrolled at half the rate with 60% of the assumed biomarker prevalence; CARBO makes the design robust to that uncertainty. Mention CARBO in Q&A territory if time permits. The bigger point: when a modern Phase I/II design can be calibrated in days instead of months, the design choice becomes about what's right for the trial, not what's feasible in the timeline. The next generation of GI oncology trials should not be limited by what a statistician can hand-tune. We have the methods to do better, and the infrastructure to deploy them.

The new GI oncology trial template



Each piece is mature methodology. The gap was infrastructure — and we're closing it.

Visual summary of the talk in four boxes. DOSE — select on biology and benefit-risk, not toxicity alone. U-BOIN and BOIN12 deliver the OBD that Project Optimus is asking for. COMBINE — when you go from one drug to two, search the joint surface. POCRM, BOIN-COMB, BLRM, not sequential lead-ins. ADAPT — monitor for resistance during the trial, not after. ctDNA, serial biopsies, biomarker-stratified subtrials. CALIBRATE — BATON makes the parameter calibration tractable so the methods become deployable. Each piece is mature methodology. The gap was never the statistics; the gap was the infrastructure to deploy what we already had. We're closing it.

Take-home

1. **The RAS era needs new kinds of trials.** Biomarker-guided, dose-optimized below MTD, resistance-aware.
2. **The methods exist.** Bayesian adaptive designs handle non-monotonic dosing, combinations, and biomarker-guided adaptation.
3. **BATON closes the deployment gap.** Days instead of months.

New RAS therapies need new kinds of trials. Methods to optimize combination doses are established. Designs that track resistance and adapt the trial in response are enrolling. What's missing is making them practical to calibrate and run.

Three things to leave with. First, the RAS era needs a new generation of trials: biomarker-guided combinations, dose-optimized below the MTD where non-monotonic dose-response makes the MTD the wrong target anyway, and resistance-aware because patients fail therapy and we need to see it happen during the trial, not after. Second, the methods to run those trials exist. Bayesian adaptive designs are the tool family — they handle non-monotonic dose-response through utility-based OBD-finding, they handle combinations through joint admissibility on toxicity and efficacy, and they handle in-trial biomarker adaptation through pre-specified rules with simulation-calibrated error control. Third, the bottleneck has been making these designs practical to deploy. BATON is our answer to that. It calibrates the parameter settings against your operating-characteristic targets in days instead of months, so design choices become about what's right for the trial rather than what's feasible in the timeline. The thesis I opened with: new RAS therapies need new kinds of trials. Methods exist. The bottleneck is making those designs practical to calibrate and run. Thank you.

Acknowledgments

Clinical collaborators

- Lisa Carey (UNC, EVOLVE PI)
- Channing Der (UNC, NEJM editorial[30])
- Jen Jen Yeh (UNC Surgery, SToP SPORE PI · NEJM editorial · PDAC collaborator)
- Ashwin Somasundaram (PANGEA PI)

EVOLVE MPI team

- Charles Perou (UNC) · Ian Krop, Eric Winer (Yale) · Antonio Wolff (Hopkins)

Methods collaborators

- Stephen Bates (MIT) · Ying Yuan (MD Anderson) · Qian Cao (FDA CDRH)

EVOLVE Statistical Working Group

- Susan Hilsenbeck (Baylor) · Nabihah Tayob (DFCI) · Ruizhe Chen (Hopkins)

Rashid Lab

- **Amber Young**, Tyler Humpherys, Dinelka Nanayakkara, Jialiu Xie, Andrew Walther

Funding

- **NCI P50-CA257911** — UNC **SToP** SPORE
- ARPA-H 140D042590009
- DOD HT9425241103100121
- NCI U01-CA274298

Der & Yeh, *NEJM* 2026 (editorial)

Acknowledgments. EVOLVE is led by Lisa Carey at UNC, with co-PIs Charles Perou at UNC, Ian Krop and Eric Winer at Yale, Antonio Wolff at Hopkins, and me. PANGEA is Ashwin Somasundaram's trial. Channing Der and Jen Jen Yeh co-authored the NEJM editorial — "Advances in RAS Therapeutics for Pancreatic Cancer," NEJM 394:1857-1861 — that accompanied the daraxonrasib monotherapy paper I leaned on for context, and Yeh is also the PI of our SPORE. On the statistical methods side, Stephen Bates at MIT and Ying Yuan at MD Anderson on the BOIN family and anytime-valid inference; Qian Cao at FDA CDRH on regulatory-grade calibration. The EVOLVE Statistical Working Group — Susan Hilsenbeck at Baylor, Nabihah Tayob at DFCI, and Ruizhe Chen at Hopkins — built out the operating characteristics for the trial. My lab, led on the BATON side by Amber Young, did the design and validation. Funding from the Selective Targeting of Pancreatic Cancer (SToP) SPORE — P50-CA257911, Yeh PI — which is the SPORE this audience represents; ARPA-H ADAPT; DOD PCARP; and an NCI U01 on pancreatic methods.

References

1. Eileen M. O'Reilly, Zev A. Wainberg, Andrew E. Hendifar, Mitesh J. Borad, Filippo Pietrantonio, Shubham Pant, Pascal Hammel, Chiara Cremolini, Gulam A. Manji, Paul E. Oberstein, Ignacio Garrido-Laguna, Christoph Springfeld, Nilofer S. Azad, Makoto Ueno, Stephen Y. Chui, Ying Zhang, Hina Patel, Yeonju Lee, Zeena Salman, Brian M. Wolpin, RASolute 302 Trial Investigators. Daraxonrasib or chemotherapy in previously treated metastatic pancreatic cancer. New England Journal of Medicine. 2026. doi:10.1056/NEJMoa2605555

2. Brian M. Wolpin, others. Daraxonrasib in previously treated advanced RAS-mutated pancreatic cancer. New England Journal of Medicine. 2026;394(18):1790–802. doi:10.1056/NEJMoa2505783

3. Yongxian Zhuang, others. The RAS(ON) multi-selective inhibitor daraxonrasib (RMC-6236) induces receptor tyrosine kinase cell-surface expression on pancreatic cancer cells. Cancer Research. 2026;86(5_Suppl_1):PR009. doi:10.1158/1538-7445.RASONCOTHER26-PR009

4. Brian M. Wolpin, Benjamin L. Musher, Gulam A. Manji, others. Daraxonrasib plus chemotherapy as first-line treatment for patients with metastatic pancreatic adenocarcinoma. AACR Annual Meeting 2026, Abstr LB407 (RMC-GI-102, NCT06445062); 2026.

5. Christophe Le Tourneau, J. Jack Lee, Lillian L. Siu. Dose escalation methods in phase I cancer clinical trials. Journal of the National Cancer Institute. 2009;101(10):708–20. doi:10.1093/jnci/djp079

6. Marie-Karelle Rivière, Christophe Le Tourneau, Xavier Paoletti, Frédéric Dubois, Sarah Zohar. Designs of drug-combination phase I trials in oncology: A systematic review of the literature. Annals of Oncology. 2015;26(4):669–74. doi:10.1093/annonc/mdu516

7. Jennifer A. Harrington, Graham M. Wheeler, Michael J. Sweeting, Adrian P. Mander, Duncan I. Jodrell. Adaptive designs for dual-agent phase I dose-escalation studies. Nature Reviews Clinical Oncology. 2013;10(5):277–88. doi:10.1038/nrclinonc.2013.35

8. Peter F. Thall, Elizabeth Garrett-Mayer, Nolan A. Wages, Susan Halabi, Ying Kuen Cheung. Current issues in dose-finding designs: A response to the US FDA oncology center of excellence project optimus. Clinical Trials. 2024;21(3):267–72.

9. Bob T. Li, Gerald S. Falchook, Gregory A. Durm, others. OA03.06 CodeBreaK 100/101: First report of safety/efficacy of sotorasib in combination with pembrolizumab or atezolizumab in advanced KRAS p.G12C NSCLC. Journal of Thoracic Oncology. 2022;17(9):S10–1.

10. U.S. Food and Drug Administration. Optimizing the dosage of human prescription drugs and biological products for the treatment of oncologic diseases. Guidance for Industry (final); 2024.

11. Amita Patnaik, others. Phase I study of pembrolizumab (MK-3475; anti–PD-1 monoclonal antibody) in patients with advanced solid tumors. Clinical Cancer Research. 2015;21(19):4286–93. doi:10.1158/1078-0432.CCR-14-2607

12. David S. Hong, others. $KRAS^{G12C}$ inhibition with sotorasib in advanced solid tumors. New England Journal of Medicine. 2020;383(13):1207–17. doi:10.1056/NEJMoa1917239

13. Maximilian J. Hochmair, Karim Vermaelen, Giannis Mountzios, others. Sotorasib (960 mg or 240 mg) once daily in patients with previously treated KRAS G12C-mutated advanced NSCLC. European Journal of Cancer. 2024;208:114204. doi:10.1016/j.ejca.2024.114204

14. Hao Zhu, Alex N. Phipps, Ying Yuan, Brad A. Davidson, Stacy S. Shord, Jiang Liu, Patricia M. LoRusso. FDA–AACR strategies for optimizing dosages for oncology drug products: Selecting dosages for first-in-human trials. Clinical Cancer Research. 2025;31(23):4872–81. doi:10.1158/1078-0432.CCR-25-0095

15. Suyu Liu, Ying Yuan. Bayesian optimal interval designs for phase I clinical trials. Journal of the Royal Statistical Society: Series C (Applied Statistics). 2015;64(3):507–23.

16. Yanhong Zhou, J. Jack Lee, Ying Yuan. A utility-based bayesian optimal interval (U-BOIN) phase I/II design to identify the optimal biological dose for targeted and immune therapies. Statistics in Medicine. 2019;38(28):5299–316.

17. Nolan A. Wages, Mark R. Conaway, John O'Quigley. Continual reassessment method for partial ordering. Biometrics. 2011;67(4):1555–63.

18. Beat Neuenschwander, Michael Branson, Thomas Gsponer. Critical aspects of the bayesian approach to phase I cancer trials. Statistics in Medicine. 2008;27(13):2420–39.

19. Sai-Hong Ignatius Ou, others. First-in-human phase I/IB dose-finding study of adagrasib (MRTX849) in patients with advanced KRASG12C solid tumors (KRYSTAL-1). Journal of Clinical Oncology. 2022;40(23):2530–8. doi:10.1200/JCO.21.02752

20. Naim U. Rashid, Xianlu L. Peng, Chong Jin, Richard A. Moffitt, Keith E. Volmar, Brian A. Belt, Roheena Z. Panni, Timothy M. Nywening, Stephen G. Herrera, Kathleen J. Moore, Sharon G. Hennessey, Ashley B. Morrison, Ryan Kawalerski, Apoorve Nayyar, Albert E. Chang, Bradley Schmidt, Hong Jin Kim, David C. Linehan, Jen Jen Yeh. Purity independent subtyping of tumors (PurIST): A clinically robust, single-sample classifier for tumor subtyping in pancreatic cancer. Clinical Cancer Research. 2020;26(1):82–92. doi:10.1158/1078-0432.CCR-19-1467

21. Steven J. Cohen, others. A phase 1b study of erlotinib in combination with gemcitabine and nab-paclitaxel in patients with previously untreated advanced pancreatic cancer. Cancer Chemotherapy and Pharmacology. 2016;77(4):693–701.

22. Ida Aronchik, others. Mechanisms of resistance to the RAS(ON) multi-selective inhibitor daraxonrasib in pancreatic cancer. ENA Symposium 2025, Revolution Medicines poster; 2025.

23. others. Mechanisms of resistance to oncogenic KRAS inhibition in pancreatic cancer. Cancer Discovery. 2024;14(11):2135–61. doi:10.1158/2159-8290.CD-24-0177

24. Mark M. Awad, others. Acquired resistance to KRAS G12C inhibition in cancer. New England Journal of Medicine. 2021;384(25):2382–93. doi:10.1056/NEJMoa2105281

25. Urszula N. Wasko, others. Tumour-selective activity of RAS-GTP inhibition in pancreatic cancer. Nature. 2024;629(8013):919–26.

26. Yi Xu, others. Tumor-intrinsic kinome landscape of pancreatic cancer reveals new therapeutic approaches. Cancer Discovery. 2025;15(2):346–62.

27. Deepak Bhamidipati, others. Impact of FDA project optimus guidance on the design of early-phase clinical trials. JCO Oncology Advances. 2025. doi:10.1200/OA-25-00084

28. Amber M. Young, others. BATON: Constrained Bayesian optimization for calibrating adaptive trials. 2026.

29. Daniel D. Von Hoff, others. Increased survival in pancreatic cancer with nab-paclitaxel plus gemcitabine. New England Journal of Medicine. 2013;369(18):1691–703. doi:10.1056/NEJMoa1304369

30. Channing J. Der, Jen Jen Yeh. Advances in RAS therapeutics for pancreatic cancer. New England Journal of Medicine. 2026;394(18):1857–61. doi:10.1056/NEJMe2600517

Questions

Thanks. Happy to take questions.